

Michelangelo Helm Chart

All 4 phases complete. Both Cadence and Temporal run as Helm subcharts in k3d. Single source of truth for NodePorts. Design doc fact-checked 5/5 by agent team.



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Overview

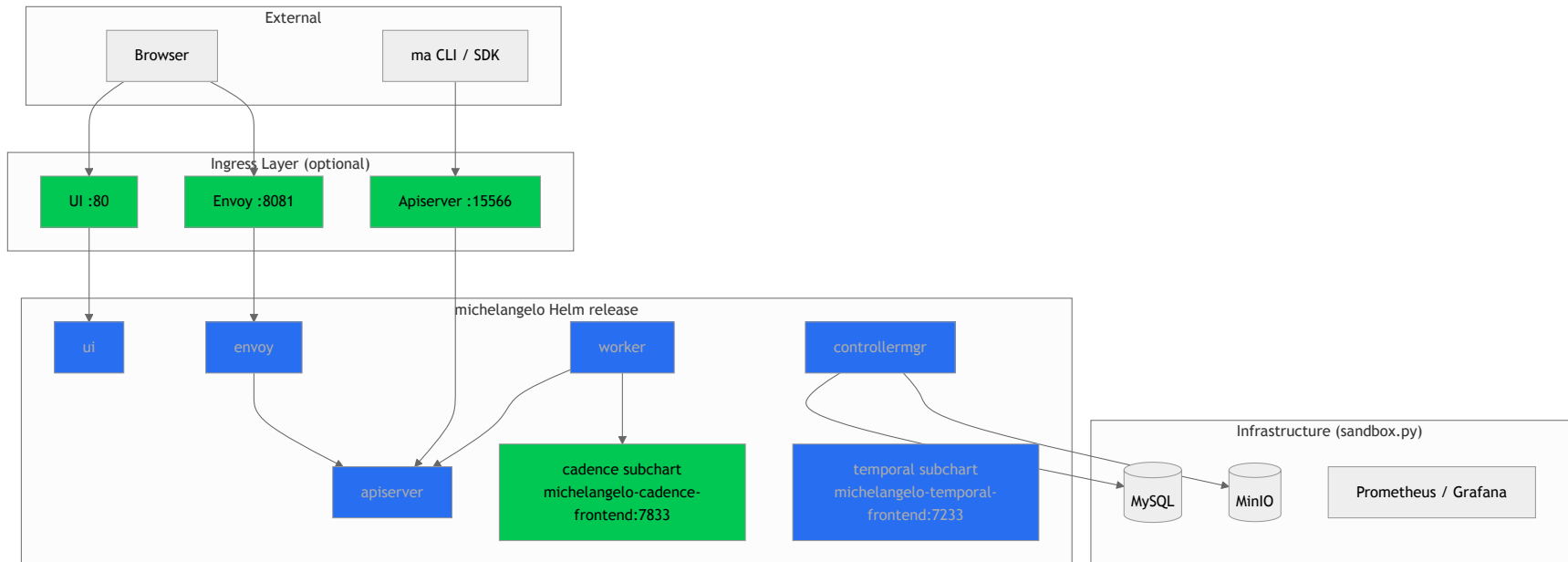
Converts the Michelangelo control plane from raw `kubectl apply` calls to a first-class Helm chart. In k3d, both Cadence and Temporal run as subcharts of the `micelangelo` release.

Chart	Installs	k3d default	Production
michelangelo	5 services + CRDs + RBAC + Ingress + validations	Always	Always
cadence SUBCHART	Cadence v1.1.0 — driver "mysql" , port 7833	Always (cadence.enabled=true)	Optional (cadence.enabled=true)
temporal SUBCHART	Temporal v0.44.0 — driver "mysql8" , port 7233	When --workflow temporal	Optional (temporal.enabled=true)

One Helm release, one workflow engine

cadence.enabled and temporal.enabled cannot both be true. templates/validations.yaml fails fast if both are set simultaneously.

Architecture



k3d Port Map — Single Source of Truth

`_helm_chart_ports(workflow)` reads NodePorts from `values-k3d.yaml` at cluster-create time. Infrastructure ports are hardcoded in `_infra_ports`. No more dual hardcoded list.

Service	Host port	NodePort	Source	Active when
MySQL	3306	30001	<code>_infra_ports</code>	Always
MinIO API	9091	30007	<code>_infra_ports</code>	Always

Service	Host port	NodePort	Source	Active when
MinIO Console	9090	30008	_infra_ports	Always
Grafana	3000	30012	_infra_ports	Always
Prometheus	9092	30015	_infra_ports	Always
Apiserver	15566	30009	values-k3d.yaml	Always
Envoy	8081	30010	values-k3d.yaml	Always
UI	8090	30011	values-k3d.yaml	Always
Cadence Web	8088	30004	values-k3d.yaml	Cadence only
Temporal Web	8080	30005	values-k3d.yaml	Temporal only
Cadence gRPC (7833) / TChannel (7933)	via port-forward	ClusterIP	Upstream chart hardcodes ClusterIP	Cadence only

Key Design Decisions

All Pods → Deployments

apiserver, envoy, worker, controllermgr promoted from bare Pods to Deployments with self-healing and rolling updates.

Schema Init Containers (two-stage)

`wait-for-metadata-storage` then `schema-init` on the `apiserver` Pod. Eliminates the ordering race condition in `sandbox.py`.

RBAC: Wildcard for `micangelo.api`

`resources: ["*"]` on the private `micangelo.api` group ensures dynamically-registered CRDs (e.g. `cachedoutputs`) are covered without chart updates. Safe — group is owned entirely by this workload. All other groups use explicit lists.

Fail-Fast Validation (`validations.yaml`)

A rendered template (not a `_helpers.tpl` file — Helm does not auto-execute top-level code in underscore files) that fails before any resource is created when `cadence.enabled && temporal.enabled` .

Per-Service `enabled` Toggle

Every service has `enabled: true` . Templates wrap each Deployment + Service + ConfigMap in `{{- if .Values.<service>.enabled }}` . The `--exclude` flag maps to `--set <service>.enabled=false` .

Cadence: cron schedule updates are a silent no-op (PR #1128)

`updateCronScheduleIfChanged()` returns `nil` on Cadence. Operators must restart trigger workflows to apply cron changes. Temporal uses `ScheduleHandle.Update()` correctly.

Cadence Subchart ALWAYS ON IN K3D

Official [cadence-workflow/cadence](https://github.com/cadence-workflow/cadence) v1.1.0. `cadence.enabled=true` in `values-k3d.yaml` — always installed as part of `helm install michelangelo` for the Cadence workflow.

MySQL driver: "mysql" (not "mysql8")

Keys: `config.persistence.database.sql.*` . Frontend: `michelangelo-cadence-frontend:7833` . Cadence Web NodePort 30004 → host 8088. Frontend stays ClusterIP (upstream chart hardcodes it); host access via `kubectl port-forward` .

Domain registration

The subchart's schema-server Job provisions MySQL databases automatically. Domain registration still runs via `kubectl run` inside the cluster — called after `_deploy_app_services()` so Cadence exists before registration is attempted.

Use helm dependency build , not update

`Chart.lock` is committed. `build` fetches the locked version; `update` re-resolves and overwrites the lock.

Temporal Subchart `--WORKFLOW TEMPORAL`

Official [temporalio/temporal](https://github.com/temporalio/temporal) v0.44.0 from <https://go.temporal.io/helm-charts> . Enabled dynamically via `--set temporal.enabled=true` when `ma sandbox create --workflow temporal` . The separate `temporaltest` Helm release is eliminated; `_setup_temporal()` removed.

MySQL driver: "mysql8" (NOT "mysql")

The Temporal chart uses a different driver string. Using `"mysql"` activates a deprecated code path. Keys: `server.config.persistence.default.sql.*` AND `server.config.persistence.visibility.sql.*` .

	Cadence	Temporal
MySQL driver	<code>"mysql"</code>	<code>"mysql8"</code>
Frontend port	7833 (gRPC)	7233
Frontend service	<code>micangelo-cadence-frontend</code>	<code>micangelo-temporal-frontend</code>
Web NodePort	30004 → host 8088	30005 → host 8080
Persistence keys	<code>config.persistence.database.sql.*</code>	<code>server.config.persistence.default.sql.*</code> + <code>visibility</code>
Schema setup	schema-server Job (auto)	schema-server Job (auto)
Domain/NS registration	Manual <code>kubectl run</code> after Helm install	Automatic (Temporal default namespace)
k3d default	Always enabled	Enabled only with <code>--workflow temporal</code>
Helm repo	<code>cadence-workflow.github.io/cadence-charts</code>	<code>go.temporal.io/helm-charts</code>

Ingress Support

Set `envoy.corsOrigins` for every Ingress install

Without it, UI loads but every API call fails with CORS. Escape dots: `--set 'envoy.corsOrigins=https://michelangelo\.example\.com'`

Phase	Path	Backend	Mode
PHASE A	/	ui:80	HTTP/1.1 — standard
PHASE A	/api	envoy:8081	gRPC-Web (HTTP/1.1 compat)
PHASE B	separate hostname	apiserver:15566	grpc (default, recommended) or passthrough

`ui.apiBaseUrl` auto-derived from `envoy.ingress.hostname` when empty and `envoy.ingress.enabled=true`.

Phase B `passthrough` mode requires `apiserver.tls.enabled=true` AND `--enable-ssl-passthrough` controller startup flag.

Migration Phases — All Complete

1 CI gate ✓

- `.github/workflows/helm-lint.yaml` triggered on `paths: ['helm/**']`
- 6 scenarios: lint, k3d profile, production Temporal, cadence subchart, temporal subchart, both-enabled fail guard

2 michelangelo chart ✓

- 34 files — 5 services as Deployments, least-privilege RBAC (`michelangelo.api: ["*"]`), schema init containers
- Cadence + Temporal optional subcharts; `validations.yaml` mutual-exclusivity guard

- Ingress Phase A (UI + Envoy, nginx path-split or subdomain) + Phase B (apiserver gRPC)

3 Boundary documentation ✓

- docs/contributing/dev-environment.md — two-tier table, service URLs, Helm dev commands

4 sandbox.py integration ✓

- helm install / upgrade replaces all control plane kubectl apply
- Cadence migrated to Helm subchart (cadence.enabled=true always in k3d); bare Pod removed
- Temporal migrated to Helm subchart (temporal.enabled=true when --workflow temporal); temporaltest release + _setup_temporal() eliminated
- _helm_chart_ports(workflow) reads all NodePorts from values-k3d.yaml — single source of truth
- Two-stage Deployment wait; minio-credentials → object-storage-credentials ; 7 YAMLS deleted
- Design doc fact-checked by agent team: 5/5 PASS

What Stays in sandbox.py

Operation	After Phase 4
k3d cluster lifecycle	Stays — requires k3d CLI

Operation	After Phase 4
MySQL pod setup	Stays — infra
MinIO pod setup	Stays — infra
Cadence	Helm subchart — <code>cadence.enabled=true</code> in <code>values-k3d.yaml</code> . Domain registration via <code>kubectl run</code> after Helm install. Frontend ports (7833/7933) via <code>kubectl port-forward</code> . Cadence Web NodePort 30004 → host 8088.
Temporal	Helm subchart — <code>temporal.enabled=true</code> via <code>--set</code> when <code>--workflow temporal</code> . Schema auto-provisioned by <code>schema-server</code> Job. <code>temporaltest release + _setup_temporal()</code> eliminated. Temporal Web NodePort 30005 → host 8080.
KubeRay, Spark Operator	Stays — separate Helm installs
Prometheus + Grafana	Stays — observability tier
MLflow, Fluent Bit	Stays — experimental tier
Compute cluster + RBAC	Stays — cross-cluster, inherently imperative
Istio + inference demo	Stays — pending inference GA

sandbox.py after Phase 4

```
_create()      → k3d create + infra + helm install michelangelo -f values-k3d.yaml
_sync()        → k3d ensure + infra sync + helm upgrade michelangelo --reuse-values
_destroy()     → helm uninstall michelangelo + infra teardown + k3d delete
_deploy_services() → removed (superseded by Helm)
```

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